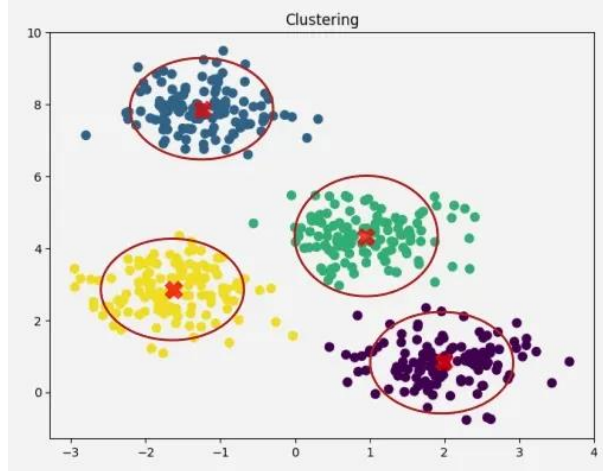


Clustering:

Clustering is an unsupervised machine learning technique used to group similar data points together without using labelled data. It helps discover hidden patterns or natural groupings in datasets by placing similar data points into the same cluster.

- Discover the natural grouping or structure in unlabeled data without predefined categories.
- Data points are assigned to clusters based on similarity or distance measures.
- Uses Euclidean distance, cosine similarity or other metrics depending on data type and clustering method.



Q1. What is Soft clustering and Hard clustering Types of Clustering

1. Hard Clustering

Hard clustering assigns each data point to exactly one cluster. A data point cannot belong to multiple clusters, making the grouping clear and easy to interpret.

- Each data point belongs to only one cluster
- No overlap between clusters
- Simple and easy to interpret

2. Soft Clustering

Soft clustering allows a data point to belong to multiple clusters with different probabilities. Instead of assigning a strict cluster, it gives a degree of membership to each cluster.

❑ Clustering Methods

1. Centroid based Clustering

- K-means: Iteratively assigns points to nearest centroid and recalculates centroids to minimize intra cluster variance.
- K-medoids: Similar to K-means but uses actual data points (medoids) as centers, robust to outliers.

2. Density based Clustering

- DBSCAN: Groups points with sufficient neighbors; labels sparse points as noise.
- OPTICS: Extends DBSCAN to handle varying densities.

3. Connectivity based Clustering

- Agglomerative: Starts with each point as a cluster and merges them step by step.
- Divisive: Starts with one cluster and splits it into smaller clusters.

4. Distribution-based Clustering

- Gaussian Mixture Model (GMM): Fits data as a weighted mixture of Gaussian distributions, assigns data points based on likelihood

5. Fuzzy Clustering

- Fuzzy C-Means: Similar to K-means but with fuzzy memberships updated iteratively

Q2. Apply single linkage clustering for the following data points = {0,9,3,6,11} and divide them into two cluster. Use Euclidean method for distance calculation. Draw dendrogram from the final cluster.

To apply **Single Linkage Hierarchical Clustering**, we repeatedly merge the two clusters having the **smallest minimum distance** between any two points.

Single Linkage Hierarchical Clustering

Given Data Points

$$\{0, 3, 6, 9, 11\}$$

Initially, every point is considered as a separate cluster.

$$C_1 = \{0\}, C_2 = \{3\}, C_3 = \{6\}, C_4 = \{9\}, C_5 = \{11\}$$

Step 1: Compute Euclidean Distance Matrix

Since these are one-dimensional data points,

$$d(a, b) = \sqrt{(a - b)^2} = |a - b|$$

Distance calculation for every pair:

d(a,b)	Calculation	Distance
0,3	$\sqrt{((0-3)^2)}=\sqrt{9}$	3
0,6	$\sqrt{((0-6)^2)}=\sqrt{36}$	6
0,9	$\sqrt{((0-9)^2)}=\sqrt{81}$	9
0,11	$\sqrt{((0-11)^2)}=\sqrt{121}$	11
3,6	$\sqrt{((3-6)^2)}=\sqrt{9}$	3
3,9	$\sqrt{((3-9)^2)}=\sqrt{36}$	6
3,11	$\sqrt{((3-11)^2)}=\sqrt{64}$	8
6,9	$\sqrt{((6-9)^2)}=\sqrt{9}$	3
6,11	$\sqrt{((6-11)^2)}=\sqrt{25}$	5
9,11	$\sqrt{((9-11)^2)}=\sqrt{4}$	2

Distance Matrix

d	0	3	6	9	11
0	0	3	6	9	11
3	3	0	3	6	8
6	6	3	0	3	5
9	9	6	3	0	2
11	11	8	5	2	0

Minimum Distance

The smallest distance is

$$d(9, 11) = 2$$

Merge

$$\{9\}, \{11\} \rightarrow \{9, 11\}$$

Current clusters:

$$\{0\}, \{3\}, \{6\}, \{9, 11\}$$

Step 2: Update Distance Matrix

Single linkage defines the distance between two clusters as where $x \in A$ and $y \in B$.

Between {9,11} and {0}

$$\min(d(9,0), d(11,0)) = \min(9, 11) = 9$$

Between {9,11} and {3}

$$\min(d(9,3), d(11,3)) = \min(6, 8) = 6$$

Between {9,11} and {6}

$$\min(d(9,6), d(11,6)) = \min(3, 5) = 3$$

Minimum Distance

There are three minimum distances:

- $d(0,3)=3$
- $d(3,6)=3$
- $d(6, \{9,11\})=3$

When a tie occurs, we merge the first pair encountered.

Merge

$$\{0\}, \{3\} \rightarrow \{0, 3\}$$

Current clusters:

$$\{0, 3\}, \{6\}, \{9, 11\}$$

Distance Matrix

d	0	3	6	{9,11}
0	0	3	6	?
3	3	0	3	?
6	6	3	0	?
{9,11}	?	?	?	0

New Distance Matrix

d	0	3	6	9,11
0	0	3	6	9
3	3	0	3	6
6	6	3	0	3
9,11	9	6	3	0

Step 3: Update Distance Matrix

Distance Calculations

Between $\{0,3\}$ and $\{6\}$

$$\min(d(0,6), d(3,6)) = \min(6,3) = 3$$

Between $\{0,3\}$ and $\{9,11\}$

$$\min(d(0,9), d(0,11), d(3,9), d(3,11)) \\ = \min(9, 11, 6, 8) = 6$$

Between $\{6\}$ and $\{9,11\}$

Already computed:

$$\min(3, 5) = 3$$

Minimum Distance

Minimum distance

$$= 3$$

Merge

$$\{0, 3\}, \{6\} \rightarrow \{0, 3, 6\}$$

Current clusters:

$$\{0, 3, 6\}, \{9, 11\}$$

Distance Matrix

d	0,3	6	9,11
0,3	0	?	?
6	?	0	3
9,11	?	3	0

New Distance Matrix

d	0,3	6	9,11
0,3	0	3	6
6	3	0	3
9,11	6	3	0

Step 4: Update Distance Matrix

Distance Calculation

Distance between

$\{0, 3, 6\}$ and $\{9, 11\}$

using Single Linkage:

$$\min \begin{cases} d(0, 9) = 9 \\ d(0, 11) = 11 \\ d(3, 9) = 6 \\ d(3, 11) = 8 \\ d(6, 9) = 3 \\ d(6, 11) = 5 \end{cases} \\ = \boxed{3}$$

Merge

$$\{0, 3, 6\}, \{9, 11\} \rightarrow \{0, 3, 6, 9, 11\}$$

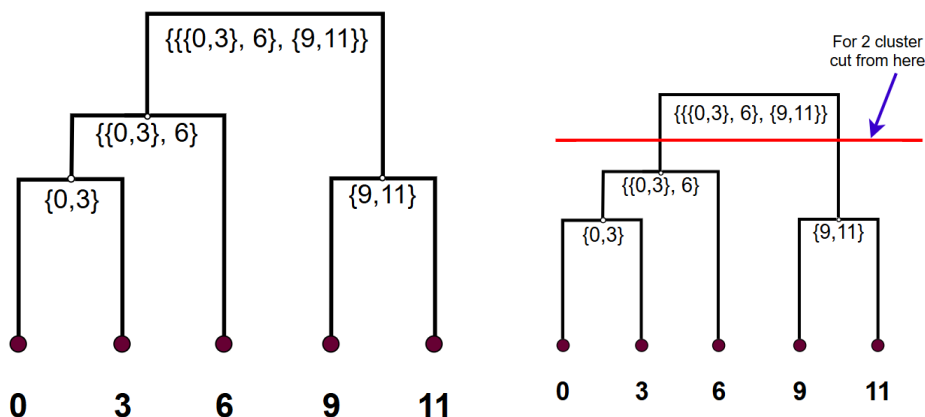
Now all points belong to one cluster.

Distance Matrix

d	0,3,6	9,11
0,3,6	0	?
9,11	?	0

New Distance Matrix

d	0,3,6	9,11
0,3,6	0	3
9,11	3	0



From this dendrogram we can derive the data points into two clusters as: $[\{0, 3, 6\}, \{9, 11\}]$

Q3. Apply complete linkage clustering for the following data points = {0,9,3,6,11} and divide them into three cluster. Use Euclidean method for distance calculation. Draw dendrogram from the final cluster.

Complete Linkage Hierarchical Clustering

To apply **Complete Linkage Hierarchical Clustering**, we repeatedly merge the two clusters having the **smallest maximum distance** between any two points.

Given Data Points

$$\{0, 3, 6, 9, 11\}$$

Initially, every point is considered as a separate cluster.

$$C_1 = \{0\}, C_2 = \{3\}, C_3 = \{6\}, C_4 = \{9\}, C_5 = \{11\}$$

Step 1: Compute Euclidean Distance Matrix

Since these are one-dimensional data points,

$$d(a, b) = \sqrt{(a - b)^2} = |a - b|$$

Distance calculation for every pair

d(a,b)	Calculation	Distance
0,3	$\sqrt{((0-3)^2)}=\sqrt{9}$	3
0,6	$\sqrt{((0-6)^2)}=\sqrt{36}$	6
0,9	$\sqrt{((0-9)^2)}=\sqrt{81}$	9
0,11	$\sqrt{((0-11)^2)}=\sqrt{121}$	11
3,6	$\sqrt{((3-6)^2)}=\sqrt{9}$	3
3,9	$\sqrt{((3-9)^2)}=\sqrt{36}$	6
3,11	$\sqrt{((3-11)^2)}=\sqrt{64}$	8
6,9	$\sqrt{((6-9)^2)}=\sqrt{9}$	3
6,11	$\sqrt{((6-11)^2)}=\sqrt{25}$	5
9,11	$\sqrt{((9-11)^2)}=\sqrt{4}$	2

Distance Matrix

d	0	3	6	9	11
0	0	3	6	9	11
3	3	0	3	6	8
6	6	3	0	3	5
9	9	6	3	0	2
11	11	8	5	2	0

Minimum Distance

The smallest distance is

$$d(9, 11) = 2$$

Merge

$$\{9\}, \{11\} \rightarrow \{9, 11\}$$

Current clusters:

$$\{0\}, \{3\}, \{6\}, \{9, 11\}$$

Step 2: Update Distance Matrix

Complete linkage defines the distance between two clusters as

$$D(A, B) = \max_{x \in A, y \in B} d(x, y)$$

Distance Calculations

Between {9,11} and {0}

$$\max(d(9,0), d(11,0)) = \max(9, 11) = 11$$

Between {9,11} and {3}

$$\max(d(9,3), d(11,3)) = \max(6, 8) = 8$$

Between {9,11} and {6}

$$\max(d(9,6), d(11,6)) = \max(3, 5) = 5$$

Minimum Distance

There are two minimum distances:

- $d(0, 3) = 3$
- $d(3, 6) = 3$

When a tie occurs, we merge the first pair encountered.

Merge

$$\{0\}, \{3\} \rightarrow \{0, 3\}$$

Current clusters: {0, 3}, {6}, {9, 11}

Distance Matrix

d	0	3	6	9,11
0	0	3	6	?
3	3	0	3	?
6	6	3	0	?
9,11	?	?	?	0

New Distance Matrix

d	0	3	6	9,11
0	0	3	6	11
3	3	0	3	8
6	6	3	0	5
9,11	11	8	5	0

Step 3: Update Distance Matrix

Distance Calculations

Between $\{0,3\}$ and $\{6\}$

$$\max(d(0,6), d(3,6)) = \max(6, 3) = 6$$

Between $\{0,3\}$ and $\{9,11\}$

$$\max(d(0,9), d(0,11), d(3,9), d(3,11)) = \max(9, 11, 6, 8) = 11$$

Between $\{6\}$ and $\{9,11\}$

Already computed:

$$\max(3, 5) = 5$$

Minimum Distance

Minimum distance

$$= 5$$

Merge

$$\{6\}, \{9, 11\} \rightarrow \{6, 9, 11\}$$

Current clusters:

$$\{0, 3\}, \{6, 9, 11\}$$

Step 4: Update Distance Matrix

Distance Calculation

Distance between

$$\{0, 3\} \text{ and } \{6, 9, 11\}$$

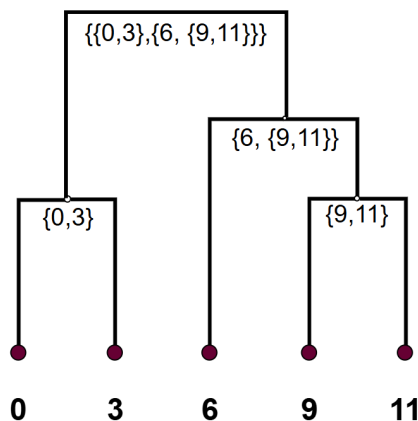
using **Complete Linkage**

$$\max \begin{cases} d(0, 6) = 6 \\ d(0, 9) = 9 \\ d(0, 11) = 11 \\ d(3, 6) = 3 \\ d(3, 9) = 6 \\ d(3, 11) = 8 \end{cases} = 11$$

Merge

$$\{0, 3\}, \{6, 9, 11\} \rightarrow \{0, 3, 6, 9, 11\}$$

Now all points belong to one cluster.



Distance Matrix

d	0,3	6	9,11
0,3	0	?	?
6	?	0	5
9,11	?	5	0

New Distance Matrix

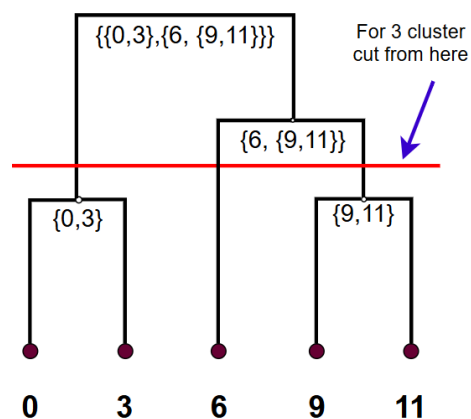
d	0,3	6	9,11
0,3	0	6	11
6	6	0	5
9,11	11	5	0

Distance Matrix

d	0,3	6,9,11
0,3	0	?
6,9,11	?	0

Distance Matrix

d	0,3	6,9,11
0,3	0	11
6,9,11	11	0



From this dendrogram we can derive the data points into three clusters as: $[\{0,3\}]$, $[\{6\}]$, $[\{9,11\}]$

Q4. Apply average linkage clustering for the following data points = {0,9,3,6,11} and divide them into three cluster. Use Euclidean method for distance calculation. Draw dendrogram from the final cluster.

Average Linkage Hierarchical Clustering

To apply **Average Linkage Hierarchical Clustering**, we repeatedly merge the two clusters having the **smallest average distance** between all pairs of points belonging to the two clusters.

Given Data Points

$$\{0, 3, 6, 9, 11\}$$

Initially, every point is considered as a separate cluster.

$$C_1 = \{0\}, C_2 = \{3\}, C_3 = \{6\}, C_4 = \{9\}, C_5 = \{11\}$$

Step 1: Compute Euclidean Distance Matrix

Since these are one-dimensional data points,

$$d(a, b) = \sqrt{(a - b)^2} = |a - b|$$

Distance calculation for every pair

d(a,b) Calculation	Distance
0,3 $\sqrt{((0-3)^2)}=\sqrt{9}$	3
0,6 $\sqrt{((0-6)^2)}=\sqrt{36}$	6
0,9 $\sqrt{((0-9)^2)}=\sqrt{81}$	9
0,11 $\sqrt{((0-11)^2)}=\sqrt{121}$	11
3,6 $\sqrt{((3-6)^2)}=\sqrt{9}$	3
3,9 $\sqrt{((3-9)^2)}=\sqrt{36}$	6
3,11 $\sqrt{((3-11)^2)}=\sqrt{64}$	8
6,9 $\sqrt{((6-9)^2)}=\sqrt{9}$	3
6,11 $\sqrt{((6-11)^2)}=\sqrt{25}$	5
9,11 $\sqrt{((9-11)^2)}=\sqrt{4}$	2

Distance Matrix

d	0	3	6	9	11
0	0	3	6	9	11
3	3	0	3	6	8
6	6	3	0	3	5
9	9	6	3	0	2
11	11	8	5	2	0

Minimum Distance

The smallest distance is

$$d(9, 11) = 2$$

Merge

$$\{9\}, \{11\} \rightarrow \{9, 11\}$$

Current clusters:

$$\{0\}, \{3\}, \{6\}, \{9, 11\}$$

Step 2: Update Distance Matrix

Average linkage defines the distance between two clusters as

$$D(A, B) = \frac{\sum d(x, y)}{|A| \times |B|}$$

where $x \in A$ and $y \in B$.

Distance Calculations

Between {9,11} and {0}

$$\frac{d(9, 0) + d(11, 0)}{2} = \frac{9 + 11}{2} = 10$$

Between {9,11} and {3}

$$\frac{6 + 8}{2} = 7$$

Between {9,11} and {6}

$$\frac{3 + 5}{2} = 4$$

Minimum Distance

There are two minimum distances:

- $d(0, 3) = 3$
- $d(3, 6) = 3$

When a tie occurs, we merge the first pair encountered.

Merge $\{0\}, \{3\} \rightarrow \{0, 3\}$

Distance Matrix

d	0	3	6	9,11
0	0	3	6	?
3	3	0	3	?
6	6	3	0	?
9,11	?	?	?	0

New Distance Matrix

d	0	3	6	9,11
0	0	3	6	10
3	3	0	3	7
6	6	3	0	4
9,11	10	7	4	0

Step 3: Update Distance Matrix

Distance Calculations

Between {0,3} and {6}

$$\frac{d(0,6) + d(3,6)}{2} = \frac{6 + 3}{2} = 4.5$$

Between {0,3} and {9,11}

$$\frac{d(0,9) + d(0,11) + d(3,9) + d(3,11)}{4} = \frac{9 + 11 + 6 + 8}{4} = \frac{34}{4} = 8.5$$

Between {6} and {9,11}

Already computed:

$$\frac{3 + 5}{2} = 4$$

Distance Matrix

d	0,3	6	9,11
0,3	0	?	?
6	?	0	4
9,11	?	4	0

New Distance Matrix

d	0,3	6	9,11
0,3	0	4.5	8.5
6	4.5	0	4
9,11	8.5	4	0

Minimum Distance

Minimum distance = 4

Merge

{6}, {9, 11} → {6, 9, 11}

Step 4: Update Distance Matrix

Distance Calculation

Distance between

{0, 3} and {6, 9, 11}

using **Average Linkage**

$$\frac{d(0,6) + d(0,9) + d(0,11) + d(3,6) + d(3,9) + d(3,11)}{2 \times 3} = \frac{6 + 9 + 11 + 3 + 6 + 8}{6} = \frac{43}{6} = 7.17$$

Merge

{0, 3}, {6, 9, 11} → {0, 3, 6, 9, 11}

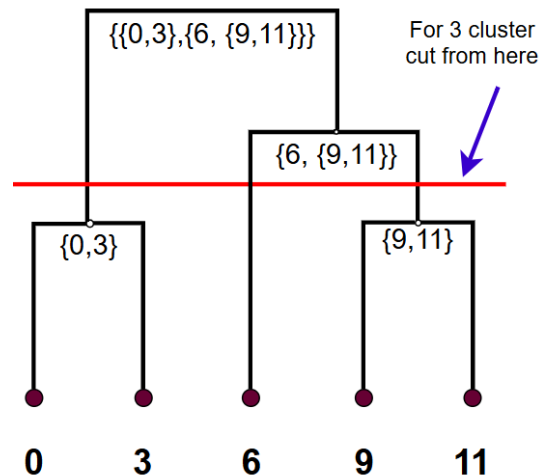
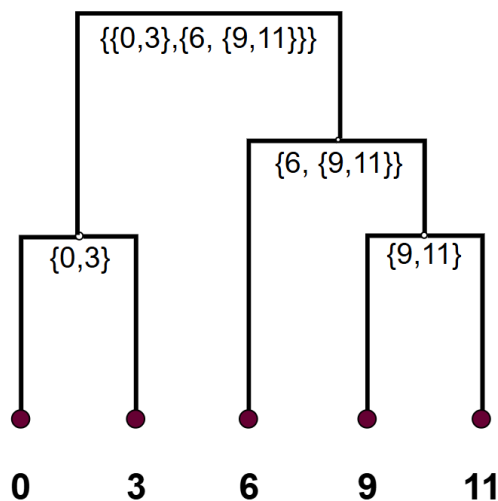
Now all points belong to one cluster.

Distance Matrix

d	0,3	6,9,11
0,3	0	?
6,9,11	?	0

New Distance Matrix

d	0,3	6,9,11
0,3	0	7.17
6,9,11	7.17	0



From this dendrogram we can derive the data points into three clusters as: [{0,3}], [{6}], [{9,11}]

Q5. Suppose that the data mining task is to cluster points into three clusters, where the points are A1(2, 10), A2(2, 5), A3(8, 4), B1(5, 8), B2(7, 5), B3(6, 4), C1(1, 2), C2(4, 9). The distance function is Euclidean distance. Suppose initially we assign A1, B1, and C1 as the center of each cluster, respectively. apply K-means clustering algorithm to find the cluster.

Given Data

Point	Coordinates
A1	(2,10)
A2	(2,5)
A3	(8,4)
B1	(5,8)
B2	(7,5)
B3	(6,4)
C1	(1,2)
C2	(4,9)

Number of clusters:

K=3

Initial centroids are

- Cluster 1: **A1 = (2,10)**
- Cluster 2: **B1 = (5,8)**
- Cluster 3: **C1 = (1,2)**

We use Euclidean distance:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Iteration 1: Current Centroids

Cluster	Centroid
C1	(2,10)
C2	(5,8)
C3	(1,2)

Step 1: Calculate distance of every point

(Values rounded to 2 decimal places.)

Point	To (2,10)	To (5,8)	To (1,2)	Assigned Cluster
A1(2,10)	0.00	3.61	8.06	C1
A2(2,5)	5.00	4.24	3.16	C3
A3(8,4)	8.49	5.00	7.28	C2
B1(5,8)	3.61	0.00	7.21	C2
B2(7,5)	7.07	3.61	6.71	C2
B3(6,4)	7.21	4.12	5.39	C2
C1(1,2)	8.06	7.21	0.00	C3
C2(4,9)	2.24	1.41	7.62	C2

Step 2: Form clusters

Cluster 1: {A1}

Cluster 2: {A3, B1, B2, B3, C2}

Cluster 3: {A2, C1}

Step 3: Compute new centroids

Cluster 1: Only A1

$$\left(\frac{2}{1}, \frac{10}{1}\right) = (2, 10)$$

Cluster 2: Points: (8,4),(5,8),(7,5),(6,4),(4,9)

$$x = \frac{8+5+7+6+4}{5} = 6, y = \frac{4+8+5+4+9}{5} = 6$$

New centroid

$$(6, 6)$$

Cluster 3: Points: (2,5),(1,2)

$$x = \frac{2+1}{2} = 1.5, y = \frac{5+2}{2} = 3.5$$

New centroid

$$(1.5, 3.5)$$

New Centroids after Iteration 1

Cluster	New Centroid
C1	(2,10)
C2	(6,6)
C3	(1.5,3.5)

Iteration 2: Use the new centroids.

Point	To (2,10)	To (6,6)	To (1.5,3.5)	Assigned Cluster
A1	0.00	5.66	6.52	C1
A2	5.00	4.12	1.58	C3
A3	8.49	2.83	6.52	C2
B1	3.61	2.24	5.70	C2
B2	7.07	1.41	5.52	C2
B3	7.21	2.00	4.53	C2
C1	8.06	6.40	1.58	C3
C2	2.24	3.61	6.04	C1

Step 4: New clusters

Cluster 1: {A1, C2}

Cluster 2: {A3, B1, B2, B3}

Cluster 3: {A2, C1}

Iteration 3

Using centroids

- (3, 9.5)
- (6.5, 5.25)
- (1.5, 3.5)

After calculating the distances again, **every point remains in the same cluster.** Therefore, no centroid changes. The algorithm **converges.**

Final Clusters

Cluster	Points	Final Centroid
Cluster 1	{A1, C2}	(3, 9.5)
Cluster 2	{A3, B1, B2, B3}	(6.5, 5.25)
Cluster 3	{A2, C1}	(1.5, 3.5)

Step 5: Compute new centroids

Cluster 1: Points (2,10),(4,9)

$$x = \frac{2+4}{2} = 3, y = \frac{10+9}{2} = 9.5$$

Centroid

(3, 9.5)

Cluster 2: Points (8,4),(5,8),(7,5),(6,4)

$$x = \frac{8+5+7+6}{4} = 6.5, y = \frac{4+8+5+4}{4} = 5.25$$

Centroid

(6.5, 5.25)

Cluster 3: Points (2,5),(1,2)

(1.5, 3.5)

(No change)

Cluster 1

- A1 (2,10)
- C2 (4,9)

Centroid = **(3, 9.5)**

Cluster 2

- A3 (8,4)
- B1 (5,8)
- B2 (7,5)
- B3 (6,4)

Centroid = **(6.5, 5.25)**

Cluster 3

- A2 (2,5)
- C1 (1,2)

Centroid = **(1.5, 3.5)**

Q6. Use K Means clustering to cluster the following data into two groups. Data Points: {2, 4, 10, 12, 3, 20, 30, 11, 25} The distance function used is Euclidean distance. Initial cluster centroid is M1 = 4 and M2 = 11.

Given Data: Data Points:

{2, 4, 10, 12, 3, 20, 30, 11, 25}

Number of clusters: $K = 2$

Initial centroids: **M1 = 4, M2 = 11**

Distance Formula: Since these are **1-dimensional data points**, the Euclidean distance becomes the absolute difference.

$$d = \sqrt{(x_2 - x_1)^2}$$

$$d = |x - c|$$

where **c** is the centroid.

Iteration 1

Current Centroids

Cluster	Centroid
C1	4
C2	11

Step 2: Form clusters

Cluster 1

{2, 4, 3}

Cluster 2

{10, 12, 20, 30, 11, 25}

Step 3: Compute new centroids

Cluster 1

$$M_1 = \frac{2 + 4 + 3}{3} = \frac{9}{3} = 3$$

Cluster 2

$$M_2 = \frac{10 + 12 + 20 + 30 + 11 + 25}{6} = \frac{108}{6} = 18$$

Iteration 2

Current Centroids

- M1 = 3
- M2 = 18

Step 1: Calculate distance of every point

Data Point	Distance to M1 = 3	Distance to M2 = 18	Assigned Cluster
2	1	16	C1
4	1	14	C1
10	7	8	C1
12	9	6	C2
3	0	15	C1
20	17	2	C2
30	27	12	C2
11	8	7	C2
25	22	7	C2

Step 1: Calculate distance of every point

Data Point	Distance to M1 = 4	Distance to M2 = 11	Assigned Cluster
2	2	9	C1
4	0	7	C1
10	6	1	C2
12	8	1	C2
3	1	8	C1
20	16	9	C2
30	26	19	C2
11	7	0	C2
25	21	14	C2

New Centroids after Iteration 1

Cluster	New Centroid
C1	3
C2	18

Step 2: Form clusters

Cluster 1

{2, 4, 10, 3}

Cluster 2

{12, 20, 30, 11, 25}

Step 3: Compute new centroids

Cluster 1

$$M_1 = \frac{2 + 4 + 10 + 3}{4} = \frac{19}{4} = 4.75$$

Cluster 2

$$M_2 = \frac{12 + 20 + 30 + 11 + 25}{5} = \frac{98}{5} = 19.6$$

New Centroids after Iteration 2

Cluster	New Centroid
C1	4.75
C2	19.6

Iteration 3

Current Centroids

- M1 = 4.75
- M2 = 19.6

Step 1: Calculate distance of every point

Data Point	Distance to M1 = 4.75	Distance to M2 = 19.6	Assigned Cluster
2	2.75	17.60	C1
4	0.75	15.60	C1
10	5.25	9.60	C1
12	7.25	7.60	C1
3	1.75	16.60	C1
20	15.25	0.40	C2
30	25.25	10.40	C2
11	6.25	8.60	C1
25	20.25	5.40	C2

Step 2: Form clusters

Cluster 1

{2, 4, 10, 12, 3, 11}

Cluster 2

{20, 25, 30}

Step 3: Compute new centroids

Cluster 1

$$M_1 = \frac{2 + 4 + 10 + 12 + 3 + 11}{6} = \frac{42}{6} = 7$$

Cluster 2

$$M_2 = \frac{20 + 25 + 30}{3} = \frac{75}{3} = 25$$

New Centroids after Iteration 3

Cluster	New Centroid
C1	7
C2	25

Iteration 4

Current Centroids

- M1 = 7
- M2 = 25

Step 1: Calculate distance of every point

Data Point	Distance to M1 = 7	Distance to M2 = 25	Assigned Cluster
2	5	23	C1
4	3	21	C1
10	3	15	C1
12	5	13	C1
3	4	22	C1
20	13	5	C2
30	23	5	C2
11	4	14	C1
25	18	0	C2

Step 2: Form clusters

Cluster 1

{2, 4, 10, 12, 3, 11}

Cluster 2

{20, 25, 30}

The clusters **do not change**, so the algorithm has **converged**.

Final Clusters

Cluster	Data Points	Final Centroid
Cluster 1	{2, 3, 4, 10, 11, 12}	7
Cluster 2	{20, 25, 30}	25

Q7. Consider the following set of data points given in Table. Cluster it using K-means algorithm with the initial value of object 2 and 5 with the coordinate values (4, 6) and (12, 4) as initial seeds.

Given Data

Object	Coordinates (X, Y)
1	(2,4)
2	(4,6)
3	(6,8)
4	(10,4)
5	(12,4)

Number of clusters: K=2

Initial centroids (given): **Object 2 = (4,6),**

Object 5 = (12,4)

We use Euclidean distance:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Iteration 1: Current Centroids

Cluster	Centroid
C1	(4,6)
C2	(12,4)

Iteration 2

Current Centroids

- C1 = (4,6)
- C2 = (11,4)

Step 1: Calculate distance of every point

Object	Coordinates	To (4,6)	To (11,4)	Assigned Cluster
1	(2,4)	2.83	9.00	C1
2	(4,6)	0.00	7.28	C1
3	(6,8)	2.83	6.40	C1
4	(10,4)	6.32	1.00	C2
5	(12,4)	8.25	1.00	C2

Final Answer

Cluster 1

Object	Coordinates
1	(2,4)
2	(4,6)
3	(6,8)

Centroid = (4,6)

Step 1: Calculate distance of every point

Object	Coordinates	To (4,6)	To (12,4)	Assigned Cluster
1	(2,4)	2.83	10.00	C1
2	(4,6)	0.00	8.25	C1
3	(6,8)	2.83	7.21	C1
4	(10,4)	6.32	2.00	C2
5	(12,4)	8.25	0.00	C2

Step 2: Form clusters

Cluster 1: {(2,4), (4,6), (6,8)}

Cluster 2: {(10,4), (12,4)}

Step 3: Compute new centroids

Cluster 1: Points: (2,4), (4,6), (6,8)

$$x = \frac{2+4+6}{3} = 4, y = \frac{4+6+8}{3} = 6$$

New centroid: (4,6)

Cluster 2: Points: (10,4), (12,4)

$$x = \frac{10+12}{2} = 11, y = \frac{4+4}{2} = 4$$

New centroid: (11,4)

New Centroids after Iteration 1

Cluster	New Centroid
C1	(4,6)
C2	(11,4)

Step 2: Form clusters

Cluster 1: {(2,4), (4,6), (6,8)}

Cluster 2: {(10,4), (12,4)}

The clusters **do not change**, so the algorithm has **converged**.

Final Clusters

Cluster	Objects	Final Centroid
Cluster 1	{1, 2, 3}	(4,6)
Cluster 2	{4, 5}	(11,4)

Cluster 2

Object	Coordinates
4	(10,4)
5	(12,4)

Centroid = (11,4)

Q8. Apply K Means clustering algorithm to cluster the following data points into two groups. Data Points: (185, 72), (170, 56), (168, 60), (179, 68), (182, 72), (188, 77). The distance function used is Euclidean distance. First two data points as initial centroids.

Given Data

Object Coordinates (Height, Weight)

P1 (185,72)
P2 (170,56)
P3 (168,60)
P4 (179,68)
P5 (182,72)
P6 (188,77)

Number of clusters: K=2

Initial centroids (first two data points):

C1 = (185,72), C2 = (170,56)

We use Euclidean distance:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Iteration 1

Current Centroids

Cluster	Centroid
C1	(185,72)
C2	(170,56)

Iteration 2

Current Centroids

- C1 = (183.5,72.25)
- C2 = (169,58)

Step 2: Form clusters

Cluster 1:

{(185, 72), (179, 68), (182, 72), (188, 77)}

Cluster 2: {(170, 56), (168, 60)}

The clusters **do not change**, so the algorithm has **converged**.

Step 1: Calculate distance of every point

Object	Coordinates	To (185,72)	To (170,56)	Assigned Cluster
P1	(185,72)	0.00	21.93	C1
P2	(170,56)	21.93	0.00	C2
P3	(168,60)	20.81	4.47	C2
P4	(179,68)	7.21	15.00	C1
P5	(182,72)	3.00	20.00	C1
P6	(188,77)	5.83	27.66	C1

Step 2: Form clusters

Cluster 1: {(185, 72), (179, 68), (182, 72), (188, 77)}

Cluster 2: {(170, 56), (168, 60)}

Step 3: Compute new centroids

Cluster 1 Points: (185,72), (179,68), (182,72), (188,77)

$$x = \frac{185+179+182+188}{4} = 183.5, y = \frac{72+68+72+77}{4} = 72.25$$

New centroid: (183.5, 72.25)

Cluster 2 Points: (170,56), (168,60)

$$x = \frac{170+168}{2} = 169, y = \frac{56+60}{2} = 58$$

New centroid: (169, 58)

New Centroids after Iteration 1

Cluster	New Centroid
C1	(183.5, 72.25)
C2	(169, 58)

Step 1: Calculate distance of every point

Object	Coordinates	To (183.5,72.25)	To (169,58)	Assigned Cluster
P1	(185,72)	1.52	21.40	C1
P2	(170,56)	21.13	2.24	C2
P3	(168,60)	19.79	2.24	C2
P4	(179,68)	6.19	14.14	C1
P5	(182,72)	1.52	19.10	C1
P6	(188,77)	6.54	27.66	C1

Final Clusters

Cluster	Data Points	Final Centroid
Cluster 1	{(185,72), (179,68), (182,72), (188,77)}	(183.5, 72.25)
Cluster 2	{(170,56), (168,60)}	(169, 58)

Q9. Apply K-Medoid clustering algorithm to form two clusters. Use Manhattan distance to find the between data point and medoid.

K-Medoids Clustering (PAM) – Step-by-Step

Solution: Distance Measure: Manhattan Distance

Given Data

Point	Coordinates (x, y)
X1	(2,6)
X2	(3,4)
X3	(3,8)
X4	(4,7)
X5	(6,2)
X6	(6,4)
X7	(7,3)
X8	(7,4)
X9	(8,5)
X10	(7,6)

We need to form **2 clusters (K = 2)**.

Point	To X1 (2,6)	To X5 (6,2)	Assigned Cluster
X1	0	8	C1
X2	3	5	C1
X3	3	9	C1
X4	3	7	C1
X5	8	0	C2
X6	6	2	C2
X7	8	2	C2
X8	7	3	C2
X9	7	5	C2
X10	5	5	C1 (tie → assign to first cluster)

Clusters after First Assignment

Cluster 1: {X1, X2, X3, X4, X10}

Cluster 2: {X5, X6, X7, X8, X9}

Step 4: Find New Medoid of Cluster 2

Cluster 2 = {X5, X6, X7, X8, X9}

Distance Table

Candidate	Sum of Distances
X5	2+2+3+5 = 12
X6	2+2+1+3 = 8
X7	2+2+3+3 = 10
X8	3+1+1+2 = 7
X9	5+3+3+2 = 13

Minimum total distance = **7**

Therefore,

New Medoid = X8 (7,4)

Step 1: Select Initial Medoids

Since the question does not specify initial medoids, we choose

- Medoid 1 = X1 = (2,6)
- Medoid 2 = X5 = (6,2)

Step 2: Calculate Manhattan Distance: Manhattan Distance Formula

$$D = |x_1 - x_2| + |y_1 - y_2|$$

Now compute the distance of every point from both medoids.

Step 3: Find New Medoid of Cluster 1

A medoid is the point having the **minimum total Manhattan distance** to all other points in the same cluster.

Cluster 1 = {X1, X2, X3, X4, X10}

Distance Table

Candidate	Sum of Distances
X1	3+3+3+5 = 14
X2	3+4+4+6 = 17
X3	3+4+2+6 = 15
X4	3+4+2+4 = 13
X10	5+6+6+4 = 21

Minimum total distance = **13**

Therefore,

New Medoid = X4 (4,7)

Step 5: Reassign Points Using New Medoids

New Medoids

- **X4 = (4,7)**
- **X8 = (7,4)**

Compute Manhattan distances again.

Point	To X4	To X8	Cluster
X1	3	7	C1
X2	4	4	C1 (tie)
X3	2	8	C1
X4	0	6	C1
X5	7	3	C2
X6	5	1	C2
X7	7	1	C2
X8	6	0	C2
X9	6	2	C2
X10	4	2	C2

Step 6: Check Whether Medoids Change Again

Cluster 1

{X1, X2, X3, X4}

Total distances

Candidate	Total
X1	9
X2	11
X3	9
X4	9

Minimum = 9 (X1, X3, and X4 tie). Since **X4** is already the current medoid, we keep X4.

Final Clusters

Cluster 1 (Medoid = X4)

Point	Coordinates
X1	(2,6)
X2	(3,4)
X3	(3,8)
X4	(4,7)

Cluster 2 (Medoid = X8)

Point	Coordinates
X5	(6,2)
X6	(6,4)
X7	(7,3)
X8	(7,4)
X9	(8,5)
X10	(7,6)

Cluster 2

{X5, X6, X7, X8, X9, X10}

Candidate	Total Distance
X5	17
X6	11
X7	13
X8	10
X9	14
X10	14

Minimum = 10

So the medoid remains X8.

Since **the medoids do not change**, the algorithm has converged.

Final Answer

- Final Medoid 1: X4 = (4,7)
- Final Medoid 2: X8 = (7,4)

Final Clusters

- Cluster 1: {X1, X2, X3, X4}
- Cluster 2: {X5, X6, X7, X8, X9, X10}

This is the final K-Medoids clustering result using Manhattan distance.

Q10. Apply K-Medoid clustering algorithm to form two clusters. Use Euclidean distance to find the distance between data point and medoid.

K-Medoids Clustering (PAM) – Step-by-Step Step 2: Euclidean Distance Formula

Solution

Distance Measure: Euclidean Distance

Given Data

Point	Coordinates (X,Y)
P1	(1,4)
P2	(5,1)
P3	(5,2)
P4	(5,4)
P5	(10,4)
P6	(25,4)
P7	(25,6)
P8	(25,7)
P9	(25,8)
P10	(29,7)

We need to form **K = 2** clusters.

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Now calculate the distance of every point from both medoids.

Point	To P1 (1,4)	To P6 (25,4)	Cluster
P1	0.00	24.00	C1
P2	5.00	20.22	C1
P3	4.47	20.10	C1
P4	4.00	20.00	C1
P5	9.00	15.00	C1
P6	24.00	0.00	C2
P7	24.08	2.00	C2
P8	24.33	3.00	C2
P9	24.74	4.00	C2
P10	28.16	5.00	C2

Step 1: Select Initial Medoids

Since the question does not specify the initial medoids, we choose

- Medoid 1 = P1 (1,4), Medoid 2 = P6 (25,4)

Step 3: Find New Medoid of Cluster 1

Cluster 1 = {P1, P2, P3, P4, P5}

Compute the total Euclidean distance from each point to all other points.

Candidate	Total Distance
P1	$5+4.47+4+9 = 22.47$
P2	$5+1+3+5.83 = 14.83$
P3	$4.47+1+2+5.39 = 12.86$
P4	$4+3+2+5 = 14.00$
P5	$9+5.83+5.39+5 = 25.22$

Minimum total distance = **12.86**

Therefore, New Medoid = P3 (5,2)

Step 5: Reassign Points Using New Medoids

New Medoids

- P3 = (5,2), P8 = (25,7)**

Calculate the Euclidean distance again.

Point	To P3	To P8	Cluster
P1	4.47	24.33	C1
P2	1.00	20.88	C1
P3	0.00	20.62	C1
P4	2.00	20.22	C1
P5	5.39	15.30	C1
P6	20.10	3.00	C2
P7	20.10	1.00	C2
P8	20.62	0.00	C2
P9	21.19	1.00	C2
P10	24.19	4.00	C2

The clusters remain unchanged.

Clusters after First Assignment

Cluster 1: {P1, P2, P3, P4, P5}

Cluster 2: {P6, P7, P8, P9, P10}

Step 4: Find New Medoid of Cluster 2

Cluster 2 = {P6, P7, P8, P9, P10}

Candidate	Total Distance
P6	$2+3+4+5.39 = 14.39$
P7	$2+1+2+4.12 = 9.12$
P8	$3+1+1+4 = 9.00$
P9	$4+2+1+4.12 = 11.12$
P10	$5.39+4.12+4+4.12 = 17.63$

Minimum total distance = **9.00**

Therefore, **New Medoid = P8 (25,7)**

Step 6: Check Medoids Again

Cluster 1: {P1, P2, P3, P4, P5}

The minimum total distance is still obtained by **P3**.

Cluster 2: {P6, P7, P8, P9, P10}

The minimum total distance is still obtained by **P8**.

Since the medoids do not change, the algorithm stops.

Final Clusters

Cluster 1 (Medoid = P3)

Point	Coordinates
P1	(1,4)
P2	(5,1)
P3	(5,2)
P4	(5,4)
P5	(10,4)

Cluster 2 (Medoid = P8)

Point	Coordinates
P6	(25,4)
P7	(25,6)
P8	(25,7)
P9	(25,8)
P10	(29,7)

This is the final K-Medoids clustering result using the **Euclidean distance** measure.

Q11. Can you explain the difference between supervised and unsupervised learning with respect to cluster analysis?

Supervised learning typically involves labeled data, in which both features and target values are provided during training. Algorithms learn to predict target values by finding patterns within this labeled data.

On the other hand, **unsupervised learning** uses unlabeled data, where only the input features are available. Algorithms operating in this mode look for hidden structure within this data, typically without any specific target in mind.

Relationship to Clustering

- **Supervised learning** tasks often do not involve cluster analysis directly. However, models trained using labeled data can sometimes be used to identify underlying clusters within the data.
- **Unsupervised learning**, and specifically **cluster analysis**, is designed to partition data into meaningful groups based solely on the provided feature set.

Q12. How does cluster analysis help in data segmentation?

Data segmentation is the process of dividing a dataset into distinct groups based on shared characteristics. **Cluster analysis** accomplishes this task and benefits various domains, from **customer segmentation** to automatic tagging in **image recognition**.

Data Segmentation and Cluster Analysis

- **Intent:** Describe the role and utility of cluster analysis in data segmentation.
- **Methods:** Examples, Visuals (if applicable)

Segmentation Example: Food Delivery

In a food delivery dataset, let's assume the goal is to segment customers based on behavioral patterns for personalized targeting.

Clusters obtained through k-means or another method could include:

- "Busy Professionals" who make frequent, small-portion orders during weekdays.
- "Occasional Foodies" who place larger orders on weekends.
- "Health Enthusiasts" who consistently order from fitness-oriented restaurants.

Segmentation Example: Image Recognition

In the context of image recognition, cluster analysis, especially through methods like k-means, can be utilized to automatically tag and organize images.

If we consider a database of wildlife images, cluster analysis can group together images of the same species based on visual features. This can be immensely useful for accurate tagging and retrieval.

Q13. What are the main challenges associated with clustering high-dimensional data?

Clustering high-dimensional data poses unique challenges that traditional methods designed for low-dimensional data may struggle to address. Let's take a look at these challenges:

Challenges of High-Dimensional Data

- **Curse of Dimensionality:** As the number of dimensions increases, the **data becomes increasingly sparse**, leading to unreliable distance metrics. This issue impairs the ability of distance-based clustering algorithms such as k-means and hierarchical clustering.
- **Degradation of Euclidean Distance:** While the Euclidean distance measure is intuitive and widely used, it often becomes less meaningful in high-dimensional spaces. The "flattening effect" makes points seem equidistant, and the influence of this effect grows with dimensionality.
- **Clustering Quality Deterioration:** High-dimensional data can result in suboptimal clustering solutions, reducing the overall quality and interpretability of clusters.
- **Loss of Discriminative Power:** With a countless number of potential projections in high-dimensional space, traditional visual inspection methods, like scatterplots, lose their effectiveness. Not all clusters are guaranteed to be adjacent in any two-dimensional projection, leading to the "small cluster" and "compact cluster" problems.
- **Increased Computational Demands:** As the feature space expands, the computational cost of clustering algorithms, particularly those dependent on pairwise distance computations, escalates significantly.

- **More Susceptibility to Noise and Outliers:** High-dimensional spaces are inherently more susceptible to noise, which can affect the validity of the cluster structure. The influence of outliers can be magnified as well.
- **Dimension Reduction Challenges:** Pre-processing high-dimensional data via dimension reduction may not always be straightforward, especially when it involves preserving certain characteristics important for clustering.
- **Interpretation and Communication Hurdles:** It is more complex to visually or conceptually convey the nature of high-dimensional clusters and their defining features.
- **Feature Selection Complexity:** In high-dimensional data, identifying which features are most relevant for the clustering task can be a challenge in itself.

Considering these challenges, selecting the most fitting clustering approach for high-dimensional datasets is crucial. Algorithms like **DBSCAN**, which are less affected by the curse of dimensionality, or density-based methods like **OPTICS** or **HDBSCAN**, are often recommended. Also, **model-based clustering methods** can show more robustness in high-dimensional settings.

Q14. Discuss the importance of scaling and normalization in cluster analysis.

Scaling and Normalization influence the outcome of a clustering analysis. Clustering is often sensitive to the scaling of the data. Therefore, it's essential to get the correct scale to ensure that the model converges accurately.

The Role of Distances

- **Euclidean and Manhattan Distances:** These metrics are sensitive to varying scales. For example, a one-unit change in a dimension with a larger scale attribute would overshadow multiple units of change in a smaller scale attribute.
- **Cosine Similarity:** This measure is more robust to scale disparities as it focuses on angles, not magnitudes.

Impact on Algorithms

- **K-Means:** This method tries to minimize the sum of squared distances within clusters. Given its use of Euclidean distance, it is sensitive to scaling.
- **Hierarchical Clustering:** The choice of distance metric (e.g., Euclidean, Manhattan, or others) influences the method's performance with scaled data.
- **DBSCAN:** This approach uses a distance parameter to identify neighbor points for core point determination. Scaled data affects this distance, thereby impacting core point identification and the clustering outcome.

Consequences of Unscaled Data

Without scaling or standardizing data, attributes whose magnitudes are orders of magnitude greater could disproportionately influence the results, leading to ineffective clusters.

Techniques for Scaling

- **Min-Max Scaling:** It transforms data within a fixed range (usually 0 to 1).
- **Z-Score Standardization:** This ensures transformed data has a mean of 0 and a standard deviation of 1.
- **Robust Scaling:** It's similar to Z-score, but uses the median and interquartile range, making it less sensitive to outliers.

Q15. How would you determine the number of clusters in a dataset?

Determining the **optimal number of clusters** in a dataset is a crucial step in most clustering techniques. Several methods can provide guidance in this regard.

Methods for Estimating Clusters

1. **Visual Inspection:** Plot the data and visually identify clusters. While this is subjective, it allows for quick insights.
2. **Elbow Method:** Compute the sum of square distances from each data point to the centroid of its assigned cluster. Plot these values for a range of cluster counts. The "elbow" point on the plot represents an optimal number of clusters, where the sum of square distances levels off.
3. **Silhouette Score:** Evaluate the quality of clusters by measuring how similar an object is to its own cluster (cohesion) compared to other clusters (separation). The silhouette score ranges from -1 to 1, with higher values indicating better-defined clusters.

4. **Gap Statistic:** Compare within-cluster variation for different numbers of clusters with that of a random data distribution. Using the "gap" between the two metrics, this method identifies an optimal number of clusters.
5. **Cross-Validation Approach:** Integrate cluster analysis with cross-validation to select the number of clusters that best fits the workflow.
6. **Information Criteria Methods:** Use statistical techniques to measure the trade-off between model fit and the number of parameters in the model.
7. **Bootstrap:** Create multiple datasets from the original one and run clustering algorithms on each. By analyzing the variability across these datasets, the "best" number of clusters can be estimated.
8. **Hierarchical Clustering Dendrogram:** Cut the tree at different heights and evaluate cluster quality to identify the optimal number of clusters.
9. **Density-Based Clustering:** Techniques such as DBSCAN do not explicitly require a predefined number of clusters. They can still provide valuable insights in terms of local neighborhood densities.
10. **Model Specific Methods:** Some clustering algorithms may have built-in methods to determine the optimal number of clusters, like the Gaussian Mixture Model through the Bayesian Information Criterion (BIC).

Q16. How does hierarchical clustering differ from K-means?

Hierarchical Clustering and the **K-Means algorithm** are both techniques for unsupervised learning, but they differ significantly in several critical aspects.

Key Distinctions

Methodology

- **K-Means** divides the dataset into pre-determined k clusters. Data points are iteratively reassigned to the nearest cluster center until little to no change occurs.
- **Hierarchical Clustering** does not require a set number of clusters. It builds a tree or dendrogram to represent the arrangement of clusters and enables different strategies for cluster extraction.

Initiation and Sensitivity

- **K-Means** is significantly influenced by the choice of initial cluster centers. The outcome may vary with different starting configurations.
- **Hierarchical Clustering** does not rely on initializations and has lower sensitivity to outliers due to its merge-divide strategy.

Execution Order

- While **K-Means** is an iterative process, initializing cluster centers at each iteration, **Hierarchical Clustering** type can utilize a "divisive" (top-down) or "agglomerative" (bottom-up) approach.
 - In agglomerative, each data point starts as its cluster and, in each step, the two closest clusters are merged till one or k clusters are left.
 - In divisive, all data points begin in one cluster, and the cluster is then successively divided into smaller, more specific clusters till a single observation or k clusters are left.

Inference Strategy

- **K-Means:** An instance is assigned to the nearest cluster center, and the overall process aims to minimize the sum of squares.
- **Hierarchical Clustering:** This method allows several ways to infer clusters. For instance, to decide the number of clusters from the dendrogram, one can choose a defined cut-off point where the vertical line passes through the tallest unbroken line.

Visual Output

- **K-Means:** Visualizing clusters can be done in 2- or 3-dimensions using scatter plots. However, the 'essence' of clusters visually extracted may vary based on the viewpoint.
- **Hierarchical Clustering:** A dendrogram is an invaluable visual representation that provides a quick overview of potential cluster counts and how individual instances group and ungroup.

Q17. What is the role of the distance metric in clustering, and how do different metrics affect the result?

The selection of an appropriate **distance metric** is vital in ensuring the quality of a **clustering algorithm**. Metrics influence the geometry of cluster shapes and can significantly impact the clustering result.

1. **Euclidean Distance:** L2 Norm is sensitive to scale and can under-perform with high-dimensional or mixed-variance data. Most widely used.

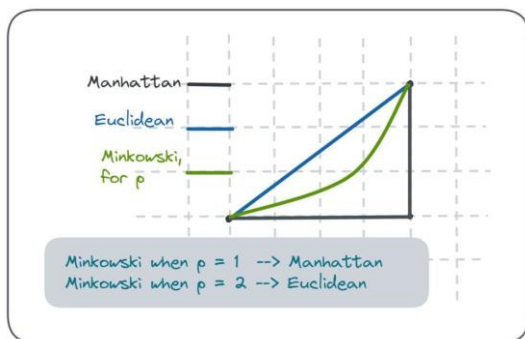
$$d(p, q) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

3. **Manhattan Distance:** Useful for highly dimensional data due to its scale-invariance. The length of a path between points is the sum of the absolute differences of their coordinates.

$$d(p, q) = \sum_{i=1}^n |q_i - p_i|$$

4. **Minkowski Distance:** When $p = 1$, this is equivalent to Manhattan distance; when $p = 2$, it's the same as Euclidean. This metric serves as a unifying framework for other distance measures.

$$d(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{\frac{1}{p}}$$



Q18. Why does K-Means fail even when you choose the correct number of clusters (K)?

Q19. Why is feature scaling critical before applying K-Means?

Q20. Which of the following algorithms is most sensitive to outliers? ans: K-means clustering algorithm

Q21. How do you choose the right number of clusters?

Q22. What is t-SNE?

Q23. Why do we normalize features in machine learning?

Q24. What's the difference between up sampling and down sampling?

Q25. What is data leakage, and how can you detect it?

Q26. What hyperparameters in Random Forest help prevent overfitting?

Q27. What is one-shot learning?

Q28. Is accuracy always a good metric for classification?

Q29. Which error metric handles outliers better: MAE, MSE, or RMSE?

Q30. List of Model Evaluation Techniques/performance measuring techniques for classification and regression

1. Performance Evaluation for Classification

Classification predicts **categorical values** (e.g., Yes/No, Spam/Not Spam).

A. Confusion Matrix

A confusion matrix summarizes prediction results.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

B. Accuracy: Percentage of correctly classified instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

C. Precision: Among predicted positives, how many are actually positive?

$$\text{Precision} = \frac{TP}{TP + FP}$$

D. Recall (Sensitivity): Among actual positives, how many are correctly identified?

$$\text{Recall} = \frac{TP}{TP + FN}$$

E. Specificity: Measures how well negatives are identified.

$$\text{Specificity} = \frac{TN}{TN + FP}$$

F. F1-Score: Harmonic mean of Precision and Recall.

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Useful when classes are imbalanced.

2. Performance Evaluation for Regression: Regression predicts continuous values

A. Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum |y - \hat{y}|$$

B. Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

C. Root Mean Squared Error (RMSE): Square root of MSE.

$$RMSE = \sqrt{MSE}$$

D. R-Squared (R^2): Measures how much variance in the target variable is explained by the model.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

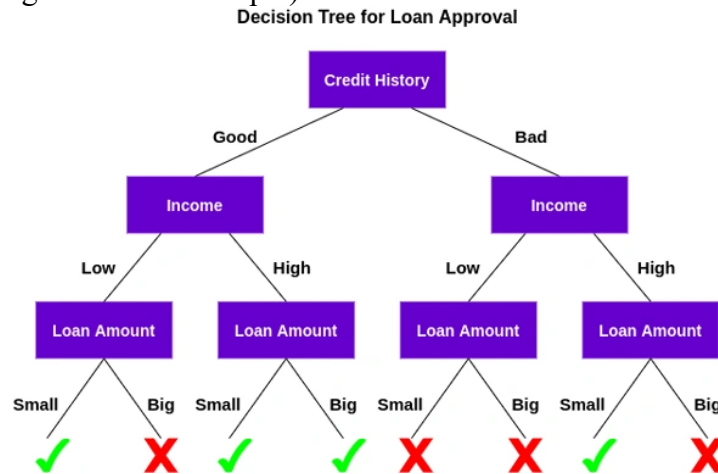
E. Mean Absolute Percentage Error (MAPE)

Measures average percentage error.

$$MAPE = \frac{100}{n} \sum \left| \frac{y - \hat{y}}{y} \right|$$

Q31: Which one is more robust to outliers between Decision Tree or Random Forest?

A decision tree is a supervised **machine-learning algorithm** that can be used for both classification and regression problems. The algorithm builds its model in the structure of a tree along with decision nodes and leaf nodes. A decision tree is a series of sequential decisions made to reach a specific result. Here's an illustration of a decision tree in action (using our above example):



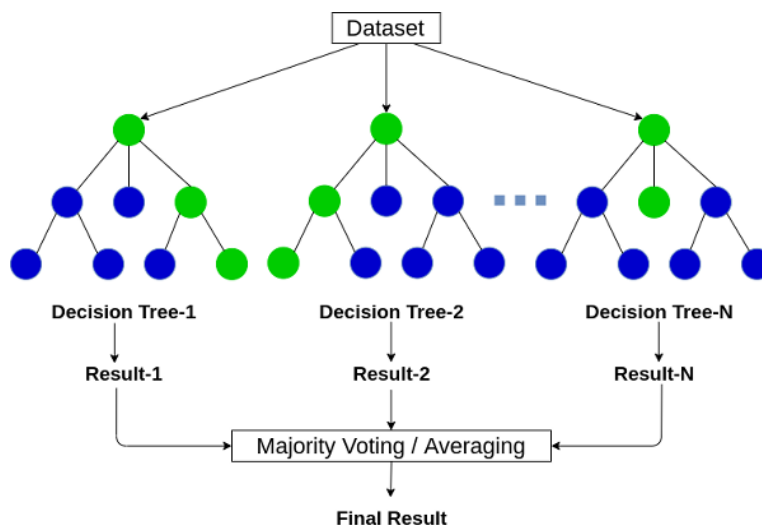
Let's understand how this tree works

First, it checks if the customer has a good credit history. Based on that, it classifies the customer into two groups: customers with good credit history and customers with bad credit history. Then, it checks the customer's income and again classifies him/her into two groups. Finally, it checks the loan amount requested by the customer. Based on the outcomes from checking these three features, the decision tree decides whether the customer's loan should be approved. The features/attributes and conditions can change based on the data and complexity of the problem, but the overall idea remains the same. So, a decision tree makes a series of decisions based on a set of features/attributes present in the data: credit history, income, and loan amount.

Random Forest is a tree-based machine-learning algorithm that leverages the power of multiple decision trees to make decisions. As the name suggests, it is a “forest” of trees!

But why do we call it a “random” forest? That's because it is a forest of randomly created decision trees. Each node in the decision tree works on a random subset of features to calculate the output. The random forest then combines the output of individual decision trees to generate the final output. Bootstrapping is the process of randomly selecting items from the training **dataset**. This is a haphazard technique. It assembles randomized decisions based on several decisions and makes the final decision based on the majority voting.

The Random Forest Algorithm combines the output of multiple (randomly created) Decision Trees to generate the final output.



This process of combining the output of multiple individual models (also known as weak learners) is called **Ensemble Learning**.

Random Forest is generally more robust to outliers than a Decision Tree.

Comparison

Feature	Decision Tree	Random Forest
Robustness to outliers	Moderate	High
Sensitivity to noisy data	Higher	Lower
Overfitting tendency	High	Lower

Decision Tree

- A single decision tree is built using the entire training dataset.
- If there are outliers or noisy samples, the tree may create unnecessary splits to fit them.
- This can lead to overfitting.

Random Forest

- A random forest consists of many decision trees.
- Each tree is trained on a bootstrap sample (random subset with replacement) of the data.
- At each split, it considers only a random subset of features.
- The final prediction is obtained by:
 - Majority voting (classification)
 - Averaging (regression)

Because an outlier appears in only some of the bootstrap samples, its influence is diluted. Averaging or majority voting further reduces the effect of any single tree that overfits to an outlier.

Example

Suppose a dataset has 1000 normal samples and 10 extreme outliers.

- Decision Tree: May create branches specifically for those 10 outliers, reducing generalization.
- Random Forest: Some trees may see the outliers, others may not. Since predictions are aggregated across many trees, the outliers have much less impact.

Random Forest is more robust to outliers because it:

- Uses bootstrap sampling,
- Builds multiple trees,
- Aggregates predictions through voting or averaging,
- Reduces the influence of noisy or extreme observations.